04

CHANGELOOP

Question 4

How much is saved, who benefits, and which figures are projections rather than results.

The question

How much water is saved per unit per year, and who benefits?

My answer

Everything below is modelled, not measured. A small number I can defend is worth more to me than a large one I cannot.

On one machine, one shift, five lots, with both human decisions approved, the model credits 1,044 litres of fresh water avoided, 62.7 kg of salt kept out of the effluent, 196 kWh of evaporator heat not burned, and ₹9,709 of cost avoided. Every delivery still goes out on time.

That comes from two separate decisions. One is choosing a better plan at the start of the shift. The other is ending wash-off when the sensors show the cloth is clean, instead of when a fixed timer runs out.

Annualised for a single machine, assuming two shifts a day across 300 working days, which is 600 comparable shifts, that works out to roughly 627,000 litres of fresh water, 37.6 tonnes of salt and 118 MWh of evaporator heat in a year.

At cluster scale the repository carries a separate projection on a per-lot basis: 400 units, 900 lots per unit per year, 360,000 lots in total. That gives about 75 million litres of fresh water, 4,500 tonnes of salt, 14,106 MWh of evaporator steam and 6,376 tonnes of CO□e annually. Those two scalings rest on different throughput assumptions, so I keep them as two separate projections rather than implying one follows from the other. Both are labelled PROJECTED in the interface, and neither should be quoted as achieved impact.

There is something else I want to put on the record, because a careful reader will find it anyway.

The 1,044 litres comes from the plan the system recommends under normal operating economics, and that plan improves water, salt and energy by about 6 percent. A much better plan exists. Switching to low-electrolyte dye chemistry cuts salt by around 46 percent, is feasible, and misses no deadline. The optimiser does not recommend it under normal economics, because the one-off chemistry premium of roughly ₹9,000 on this batch outweighs a single shift's savings. Price water scarcity or carbon higher, which is what happens under a drought restriction, and the same engine starts choosing it.

> The daily scheduling decision delivers the modest number. The large number is a procurement decision the system can justify but cannot make on its own.

Who benefits. The dyeing unit, through a lower treatment charge and less salt bought. The common effluent treatment plant, through less steam purchased and less salt it has nowhere to put. And the basin, through lower freshwater abstraction.

I am not claiming a number of people served or hectares protected. I do not have evidence strong enough to defend a figure like that, and making one up would undermine everything else here.

The first real impact measurement comes from a pilot, starting with four weeks of baseline metering before any recommendation is allowed to change what the plant does.

Supporting notes

Why the answer is shaped this way

Impact questions invite inflation, and an inflated number is the easiest thing for a panel to puncture. So the answer opens by labelling everything a projection and closes by refusing to estimate people served.

The 6 percent against 46 percent paragraph is deliberate. A judge who reads the repository will discover that the recommended plan under normal operation saves 6 percent while a far better plan sits unused. Found by them, it reads as an overclaim being caught. Said first, it reads as understanding your own economics. It is also a genuinely interesting output: the model is showing where the payback threshold for the chemistry switch actually sits.

Every figure, and how to check it

Figure	Value	Source
Freshwater avoided, one shift	1,044.2 L	`figures.json` → `impact.fresh_av`
Salt avoided, one shift	62.65 kg	`impact.salt_av`
Evaporator heat avoided, one shift	196.4 kWh	`impact.mee_av`
Cost avoided, one shift	₹9,709	`impact.cost_av`
Deliveries late in recommended plan	0	`options.OPTION_B.late`
Annualisation assumption	2 shifts/day × 300 days = 600	stated assumption, not measured
Annual, one machine	~627,000 L, 37.6 t, 118 MWh	600 × the per-shift figures
Cluster projection	75.0 M L, 4,500 t, 14,106 MWh, 6,376 t CO□e	`figures.json` → `cluster`
Cluster assumptions	400 units × 900 lots/unit/yr, 12.5 kg salt/lot	`core/economics.py` → `cluster_projection()`
Recommended plan improvement	6.0% on water, salt and energy	`options.OPTION_A` against `OPTION_B`
Best feasible low-salt plan	382.1 kg salt against 714.4 baseline, −46%, 0 h late	exhaustive search across all 480 candidates
Chemistry premium on this batch	~₹9,042	`low_salt_chemistry_cost_inr_per_kg_fabric` × fabric weight

A correction made while preparing this answer

The cluster projection used to report 398.9 million litres. That figure was wrong.

It treated freshwater as an independent input of 1,108 litres per lot, while deriving reject volume from salt. The two then disagreed by 5.3× inside the same block of output: 398,880 m³ of freshwater against 75,000 m³ of reject.

In a closed loop those are the same water. Makeup equals evaporative loss equals reject volume. The engine enforces that identity everywhere else and reports both as exactly 1,044.2 L on the reference shift. The cluster projection was the one place that did not, and it happened to be the place carrying the headline number.

Freshwater is now derived from reject so the two cannot drift apart again, and three tests lock it. The corrected figure is 75.0 million litres. Salt, steam and CO□e were already derived from salt and did not change.

This is in the answer because the correction cut my own headline number by more than five times. Better that the panel sees me audit my own figures than that they find it themselves.

What the answer deliberately avoids claiming

- No people served, no hectares protected, no lives improved.
- No suggestion the cluster figure is achievable. Every unit resembling the modelled one, and every recommendation being approved, are both untested.
- No merging of the two scaling bases into a single larger total.

The hardest follow-up

"Six percent isn't very much."

It is not, and that is the honest figure for one scheduling decision under normal economics. Two things still make it worth having. It costs almost nothing, because there is no new equipment involved, only a different running order. And the same engine shows exactly when the larger 46 percent move becomes worth buying, which is the decision a unit actually needs help with. Something that tells you a lever is not worth pulling today is more useful than something that always says yes.